

3. Gas Laws

The equation that says how hard it pushes

Heat a trapped gas and the pressure rises in proportion

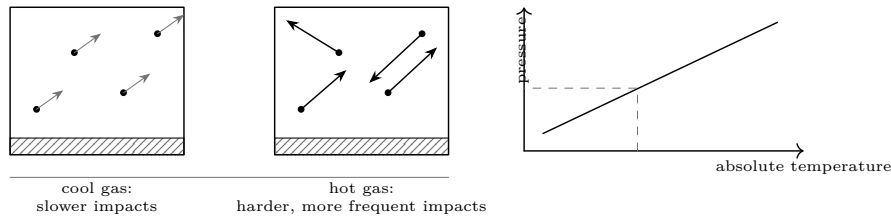
Lesson 3 · about 75 minutes · everyone — kitchen care only

What you need

- A balloon, a bottle, a bowl of hot water and one of iced water
- A pressure gauge and a bicycle pump
- A syringe with the nozzle blocked, and weights
- A thermometer
- A marshmallow and a syringe, for entertainment

Predict first

A sealed rigid container of air at room temperature is heated from 20°C to 1000°C , which is roughly what a burning fuel-air mixture reaches. The volume cannot change. By what factor does the pressure rise? Guess before you calculate — then calculate.



Do it

1. **Balloon on a bottle.** Stretch a balloon over an empty bottle's neck. Stand the bottle in hot water — the balloon inflates. Then iced water — it is drawn in. Nothing entered or left.
2. **Squeeze a syringe.** Block the nozzle, push the plunger, and feel the resistance rise. Halve the volume and the force roughly doubles. That is Boyle's law under your thumb.
3. **Marshmallow.** A marshmallow in a blocked syringe. Pull the plunger and it swells; push and it shrinks. The gas trapped in its foam is obeying the same rule.
4. **Measure safely.** Use only a purpose-made classroom gas-law apparatus within its instructions. Never heat a tyre, improvised bottle, PVC chamber, or sealed unknown vessel.
5. **Do model arithmetic.** Calculate a hypothetical fixed-volume ratio from supplied temperatures. Label it "idealized prediction," not "launcher pressure."

What it means

The gas law	$PV = nRT$. Pressure times volume equals amount times temperature, with a constant. Fix the volume and the amount, and pressure is simply proportional to <i>absolute</i> temperature.
Kelvin, not Celsius	The proportionality only works from absolute zero. Going from 20 °C to 40 °C is not a doubling — it is 293 K to 313 K, about 7%. This is the mistake everyone makes once.
There is no launcher number here	Combustion changes temperature, composition, heat transfer, volume, and time together. Two guessed temperatures cannot establish peak pressure or a safety margin.
And n changes too	Burning also changes the number of gas molecules. That is a second, smaller contribution on top of the temperature effect.
No inferred margin	A water-service pressure marking screens parts; it is not combustion approval. Do not compare it with a guessed transient and announce a safety factor.
Force, not pressure, moves the potato	Pressure times area. Three extra atmospheres on a 1.5 in bore is roughly 80 lb pushing on the potato's base, for the whole length of the barrel.

Break it

Longer barrel	A model predicts competing effects from work, expansion, and friction. Explore them in algebra or a safe bench model; do not build alternate barrels to optimize a weapon.
Bigger bore	More area, so more force — but also more gas volume needed to fill it, so the pressure falls faster as the potato moves. Argue both sides before you assume bigger is better.
Cold day	Starting at 0 °C instead of 20 °C is only a 7% difference in the gas law — but it makes PVC substantially more brittle, which matters far more. <i>see the Safety sheet</i>
Work it out first	Energy delivered is roughly average force times barrel length. Compare that with $\frac{1}{2}mv^2$ for a 200 g potato and predict the muzzle velocity you will measure in Lesson 4.

The link to the launcher

Engineering means stating what a model can and cannot support. Here you can calculate an ideal relationship, measure a purpose-made classroom apparatus, and quantify the residual. You cannot infer a combustion peak, component approval, or expected launcher velocity from those results.

Data and acceptance

Trial	T_1 (K)	P_1	T_2 (K)	P_2 measured / predicted / residual
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1
2
3

Troubleshoot

If points curve, ask whether volume changed, the apparatus leaked, the sensor lagged, or temperatures were unequal. Never hide an outlier without recording why.

ACCEPTANCE GATE

Pass only with units, kelvin temperatures, discussed residuals, and no claim that the experiment certifies a chamber.

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